



Machine learning-based modeling to predict inhibitors of acetylcholinesterase

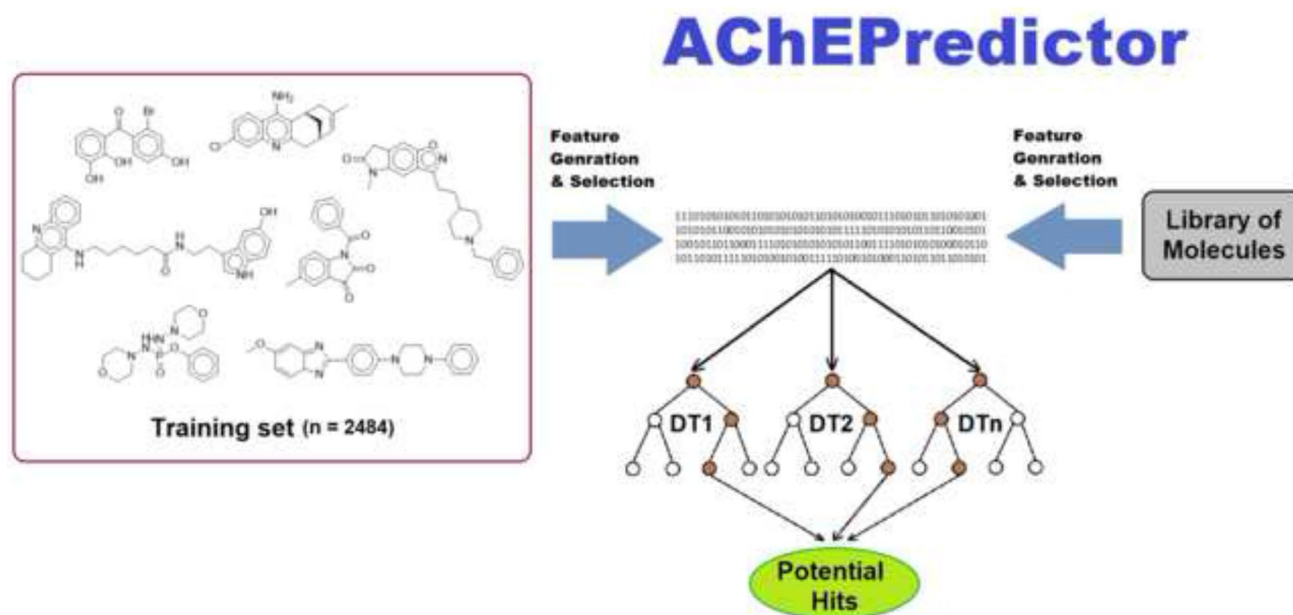
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Abstract

Acetylcholinesterase enzyme is responsible for the degradation of acetylcholine and is an important drug target for the treatment of Alzheimer's disease. When this enzyme is inhibited, more acetylcholine is available in the synaptic cleft for the use, which leads to enhanced memory and cognitive ability. The aim of the present work is to create machine learning models for distinguishing between AChE inhibitors and non-inhibitors using algorithms like support vector machine (SVM), *k*-nearest neighbor (*k*-NN) and random forest (RF). The developed models were evaluated by 10-fold cross-validation and external dataset. Descriptor analysis was performed to identify most important features for the activity of molecules. Descriptors which were identified as important include maxssCH2, minHssNH, SaasC, minssCH2, bit 128 MACCS key, bit 104 MACCS key, bit 24 estate fingerprint and bit 18 estate fingerprints. The model developed using fingerprints based on random forest algorithm produced better results compared to other models. The overall accuracy of best model on test set was 85.38 percent. The developed model is available at <http://14.139.57.41/achepredictor/>.

Graphic abstract



Keywords Acetylcholinesterase (AChE) · *K*-nearest neighbor (*k*-NN) · Support vector machine (SVM) · Random forest (RF) · Machine learning · Shiny application

Introduction

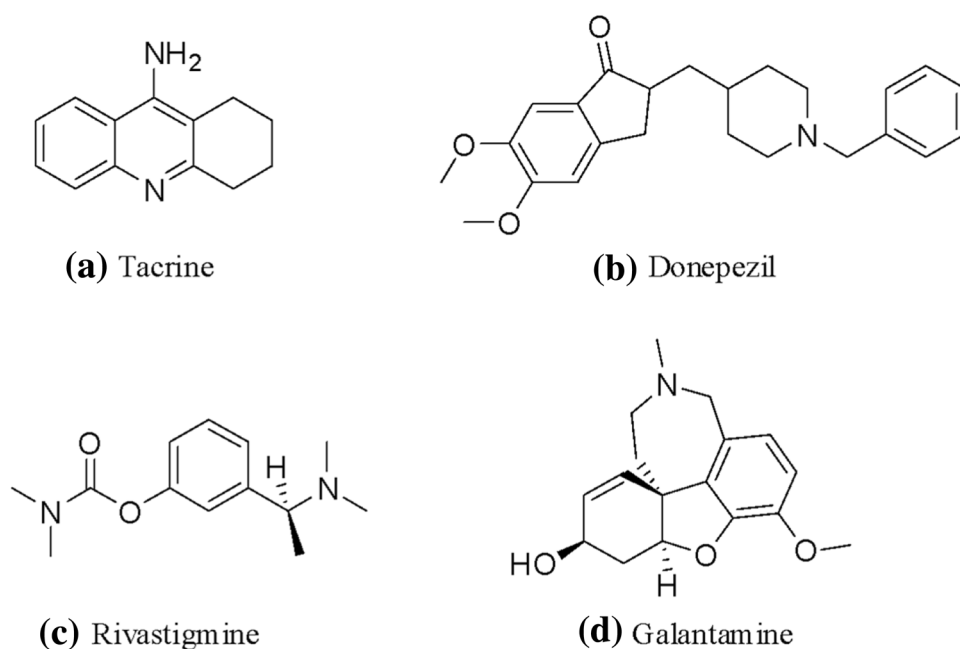
Alzheimer's disease (AD) is the most common cause of clinical dementia which is caused due to alteration of amyloid beta cascade protein and hyperphosphorylation tau protein [1]. It is a chronic progressive neurodegenerative disease which affects more than 10% of elderly people over 65 years of age [2]. It affects 35 million people all over the world and it can rise up to 115 million by 2050 [3]. Symptoms include with early sign of loss of memory, cognitive impairment and decline inability to use language. The cause of AD is poorly understood and various risk factors include genes, age, depression, and hypertension. As per the cholinergic hypothesis, there is a reduction in the synthesis of neurotransmitter acetylcholine in the individuals suffering from AD [4]. In the synaptic cleft, acetylcholine is converted into choline and acetate by the action of acetylcholinesterase (AChE) enzyme. Therefore, inhibition of AChE activity is useful for treatment for AD, thereby increasing the levels of acetylcholine for more communication between the brain cells [5]. In this way, acetylcholinesterase-inhibiting drugs are useful as symptomatic treatment (improves cognitive function and also prevents the disease onset). Only four AChE inhibitors are approved for the treatment of AD, i.e., Tacrine, Donepezil, Galantamine and Rivastigmine (their 2D structures are given in Fig. 1).

Discovery of new AChE inhibitors using the experimental techniques is costly, time-consuming and labor intensive. Therefore, it is necessary to develop the computational methods for the detection of potential lead compounds. Number of techniques like QSAR, molecular docking and

drug repurposing have been used by scientific community to identify lead molecules as AChE inhibitors [6–8]. Pascoini et al. used fragments-based de novo drug design in combination with molecular docking for designing novel inhibitors of AChE [9]. In recent years, numerous studies have shown that machine learning can be very useful for screening of lead molecules [10–13]. To our knowledge, there are only two published work for classification of AChE inhibitors using machine learning. Xue and Lv have used a dataset of 462 molecules and developed various machine learning methods like SVM, *k*-NN and C4.5 decision tree (C4.5 DT) for predicting AChE inhibitors. The prediction accuracies of their best model SVM are 88% for AChEIs and 79.6% for non-AChEIs [14]. Yang et al. have built classification models using 515 molecules based on SVM, artificial neural networks (ANN) and *k*-NN with the average prediction accuracy of the best model (SVM) ranging from 87.3 to 92.7% [15]. These previously published studies have used very small dataset of molecules. Also, the authors did not give any option to utilize the developed models.

In this work, a much larger dataset is used for development of predictive classification models. Various machine learning algorithms, i.e., SVM, *k*-NN, and RF are used for the development of models based upon molecular descriptors and fingerprints calculated using PaDEL and rcdk, respectively [16, 17]. Firstly, recursive feature elimination and Boruta are used as feature selection methods to obtain the relevant molecular descriptors for reducing dimensionality and improving the predictive performance of models. Further, we have done descriptor analysis using the cutpoint package in RStudio in order to know the most important

Fig. 1 FDA approved Acetylcholinesterase Inhibitors for Alzheimer's disease



descriptors which are able to differentiate between inhibitors and non-inhibitors.

Materials and methods

Dataset

A dataset of total 5692 molecules for which inhibitory activity against human acetylcholinesterase is available was downloaded from BindingDB database [18]. The collected data were subjected to refinement in order to remove duplicate molecules, salt forms, heavy metal and fragments. Finally, out of 5692 molecules 4140 unique molecules were left. For class labelling, IC₅₀ cutoff value of 1000 nM was decided. Molecules which were having IC₅₀ less than or equal to 1000 nM were selected as positive or inhibitors. Molecules which were having IC₅₀ value greater than 1000 nM were selected as negative or non-inhibitors. In this way, out of 4140 molecules 1862 molecules were selected as inhibitors and 2278 molecules were selected as non-inhibitors.

Descriptor generation and selection

Molecular descriptors are used to represent the physico-chemical properties of molecules, which are mainly used in SAR and QSAR studies. Total 1444 descriptors (1D and 2D descriptors) were calculated by using PaDEL software (version 2.21). Only 1D and 2D descriptors were calculated so as to avoid complications related to minimization of structure of molecules for generation of 3D descriptors. Also, 245-bit fingerprints (166-bit MACCS key and 79-bit estate fingerprints) were calculated using the rcdk package (wrapper of CDK for R). After that, the dataset was divided into the training, test and external set in 3:1:1 ratio (2484, 828 and 828 in training, test and external set, respectively), followed by the normalization using ‘range’ method to bring them in range of 0 to 1 using preprocess function of caret package available for R language. Descriptors with zero variance were removed from the dataset because they do not have any significance for the desired property of the molecules. After that, highly correlated features with correlation coefficient greater than 0.70 were removed from the dataset to avoid multicollinearity. Descriptor/feature selection was performed with the help of RFE and Boruta (feature selection and normalization were not performed on fingerprints).

- *Recursive Feature Elimination (RFE)*: This algorithm iteratively builds models with reduced number of features in each iteration and ranks the input features according to their importance for model building.

- *Boruta*: It works as a wrapper algorithm around random forest for determining the importance of each feature and gives a set of all relevant features. It generates shadow features by randomizing the values of the original feature. After each run of random forest, the importance score of features is calculated. If the importance of a shadow feature is higher than or equal to the importance of the original feature then it is designated as irrelevant.

Only train set was used in all the preprocessing steps, i.e., removal of constant features, removal of highly correlated features and normalization for feature selection to avoid leakage of information from test set to the developed model. This helps in avoiding overfitting and positively biased results on the test set. Train set was used for model fitting and optimization of hyperparameters with the help of 10-fold cross-validation and the test set was used for model selection.

Machine learning algorithms

k-nearest neighbor (*k*-NN)

k-NN is a simple supervised machine learning algorithm proposed by Cover and Hart in the year 1967 for performing the pattern classification task [19]. It classifies query data points based on majority vote count among *k* nearest data points or nearest neighbors. The different types of distances like Euclidean Distance, hamming distance and Cosine distance, etc. can be used to find the nearest neighbors.

Support vector machine (SVM)

SVM is statistical supervised machine learning algorithm originally developed by Boser and Vapnik, which is useful for recognizing refined patterns in complex datasets [20]. SVM creates a hyperplane or set of hyperplanes in a multidimensional space, which can be used for classification, regression, and other purposes. A good separation among the classes is attained by the hyperplane that is farthest from support vectors or nearest training data point of any class in high-dimensional space; the larger the margin the lesser will be the generalization error of the classifier [21].

Random forest (RF)

RF is an ensemble of a large number of decision trees. It is used mainly for classification and nonlinear regression models. In this algorithm, subsets of features are randomly sampled from the training data and each subset is used for constructing a decision tree. Finally, an ensemble decision is taken from all the constructed decision trees for providing output [22].

Model validation

In this study, we have used 10-fold cross-validation and external dataset for validation of models. A number of statistical parameters, i.e., sensitivity, specificity, accuracy, f1 score and Matthews's Correlation Coefficient (MCC) are used to evaluate the performance of the models. The percentage of inhibitors, which are correctly predicted by the model, is called sensitivity. While, the percentage of non-inhibitors, which are correctly predicted by the model, is called specificity, and accuracy is defined as a measure of the proportion of predicting the data points correctly. MCC measures the quality of binary classification. MCC value ranges from -1 to 1, if the value is closer to 1 that means the model is highly accurate and predictions are positively correlated with the actual values. If value is closer to -1 that means there is negative correlation between prediction and actual value and if value is closer to 0 that means the developed model is equivalent to random selection of class labels.

Results and discussion

A total of 179 descriptors and 116 fingerprints were left after removing constant and correlated descriptors. RFE was performed on descriptors with the help of implementation given in caret package of R. The variable scores from the RFE method were obtained using the varImp function of caret package. A Boruta run of 100 iterations was performed with the help of Boruta package. The variable importance scores of all 100 iterations were averaged. Then, the top 30 features based on variable importance score from both the methods were selected for further modeling. The features were divided into three sets as follows:

- Set-A: top 30 features from RFE method.
- Set-B: top 30 features from Boruta method.
- Set-C: 116-bit fingerprint.

Using each set of selected features, models were developed with the help of RF, *k*-NN and SVM implementations given in caret package. For SVM (radial kernel), *cost* (values ranging from 0.1 to 5.0 with a step size of 0.1) and *sigma* (values ranging from 0.05 to 2.00 with a step size of 0.05) values were optimized for obtaining the best fit using grid search method given in caret package, i.e., tuneGrid. For RF models, *mtry* value (2, 5, 8... 24, 27 and 30) was optimized. For *k*-NN models, *k* value (3, 5, 7, 9, 11 and 15) was optimized for all the three sets of features.

The results of the models developed (with optimized parameters) using the features selected with the help of RFE method are given in Table 1. From Table 1, it is clear that all three algorithms are showing similar results with slight changes in statistical parameters. But overall, the SVM (*C* = 2.4 and *sigma* = 0.1) model was showing the best results. SVM is better from *k*-NN and RF in terms of both test set and cross-validation results because of better and balanced sensitivity and specificity values. Results for the models developed using set-B features are given in Table 2. Set-B models are also showing more or less similar results. But in comparison RF model (*mtry* = 5) is better than both SVM and *k*-NN models in terms of accuracy and also because of comparatively balanced sensitivity and specificity values. Hence, among set B models RF (*mtry* = 5) was selected as the best model. In Table 3, the results of models developed with 116-bit fingerprints are given. As can be seen from Table 3, RF model (*mtry* = 13) is performing better than *k*-NN and SVM models. Therefore, RF was selected as the best one out of models developed using fingerprints or set C features. The best models from set A, set B and set C were compared to select one overall best model. Models from set A and set C performed better than set B model and are almost equivalent to each other. In the case of 10-fold cross-validation, RF (set C) performed

Table 1 Results of models trained using set A features

S No.	Model		Accuracy (%)	Sensitivity (%)	Specificity (%)	MCC	F1 score
1	RF (<i>mtry</i> = 5)	10-fold CV	84.08	77.50	89.45	0.67	0.81
		Train	99.63	99.82	99.48	0.99	0.99
		Test	84.42	78.72	88.84	0.68	0.81
2	<i>k</i> -NN (<i>k</i> = 3)	10-fold CV	81.60	80.88	82.18	0.62	0.79
		Train	89.93	88.98	90.70	0.79	0.88
		Test	82.60	82.32	82.83	0.64	0.80
3	SVM (<i>C</i> = 2.4 & <i>sigma</i> = 0.1)	10-fold CV	83.61	80.44	86.20	0.66	0.81
		Train	95.28	95.70	94.95	0.90	0.94
		Test	85.86	83.42	87.76	0.71	0.83

Highlighted values indicate the models selected

Table 2 Results of models trained using set B features

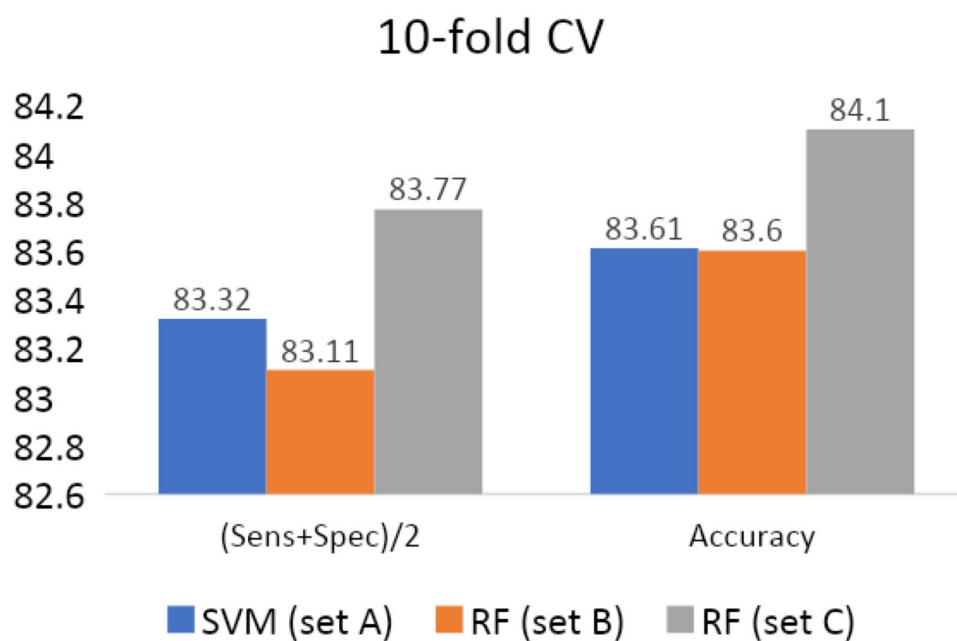
S No.	Model		Accuracy (%)	Sensitivity (%)	Specificity (%)	MCC	F1 score
1	RF (mtry = 5)	10-fold CV	83.60	78.22	88.00	0.66	0.81
		Train	99.63	99.73	99.56	0.99	0.99
		Test	85.14	79.83	89.27	0.69	0.82
2	<i>k</i> -NN (<i>k</i> = 3)	10-fold CV	81.60	80.88	82.18	0.62	0.79
		Train	89.93	88.98	90.70	0.79	0.88
		Test	81.76	81.55	82.04	0.63	0.79
3	SVM (<i>C</i> = 4.2 & sigma = 0.2)	10-fold CV	82.89	77.33	87.45	0.65	0.80
		Train	97.30	98.11	96.63	0.94	0.97
		Test	83.81	80.66	86.26	0.67	0.81

Highlighted values indicate the models selected

Table 3 Results of models trained using set C features (116-bit fingerprints)

S No.	Model		Accuracy (%)	Sensitivity (%)	Specificity (%)	MCC	F1 score
1	RF (mtry = 13)	10-fold CV	84.10	80.44	87.09	0.67	0.82
		Train	96.09	95.88	96.26	0.92	0.95
		Test	85.38	84.25	86.26	0.70	0.83
2	<i>k</i> -NN (<i>k</i> = 3)	10-fold CV	81.50	84.18	79.31	0.63	0.80
		Train	89.04	90.06	88.22	0.78	0.88
		Test	81.88	85.35	79.18	0.64	0.80
3	SVM (<i>C</i> = 2 & sigma = 0.05)	10-fold CV	81.71	73.77	88.20	0.63	0.78
		Train	97.02	96.95	97.07	0.93	0.96
		Test	81.76	74.58	87.33	0.62	0.78

Highlighted values indicate the models selected

Fig. 2 Bar plot of 10-fold cross-validation results of best models chosen from set A, B and C features

better than SVM (set A) (Fig. 2). For test set, results SVM (set A) has higher accuracy (Fig. 3) but sensitivity and specificity values of RF (set C) were more balanced. It will not make much of a difference if one from set A or

set C model is selected as the best model. Still, random forest (mtry = 5) which was trained using set C features was chosen as the final model because of slightly better performance in 10-fold cross-validation.

Prediction on external dataset

An external dataset consisting of 828 molecules was kept untouched throughout the development phase of models. External dataset serves as an important tool for validation of developed models. External dataset helps in evaluating the robustness and generalizability of developed models. Results of external set prediction are given in Table 4. As can be seen from Table 5, all three models are showing good results on external set reaffirming the robustness of developed models. RF model trained using set C features (fingerprints) is showing the best performance on external set with highest accuracy and sensitivity. The ROC curve of the best model (RF with set C features) for external set prediction is given in Fig. 4. The area under the ROC

curve for this model is 0.917. Thus, it can be said that the selected model is indeed showing good performance on the external set.

Applicability domain

Tanimoto similarity metric was used for defining the applicability domain of query molecules. If a query molecule was having Tanimoto similarity greater than or equal to 0.7 to at least one of the training set compounds, then the molecule was considered within the applicability domain. The Tanimoto similarity was calculated using the 116-bit fingerprints which were used for training the models. Using this criterion, we used the final model for prediction on molecules of external set which were within the applicability domain.

Fig. 3 Bar plot of test set results of best models chosen from set A, B and C features

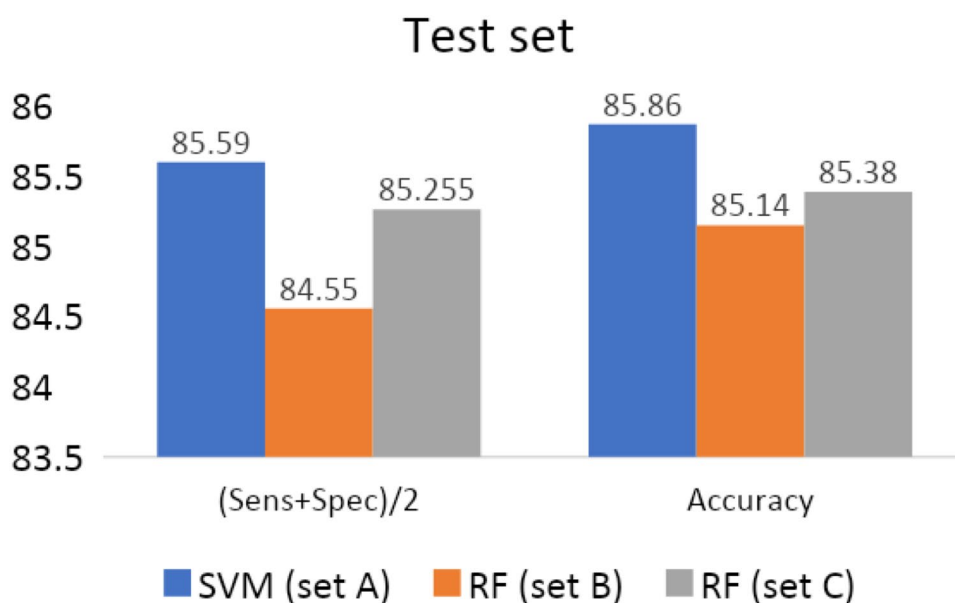


Table 4 Results of external set prediction

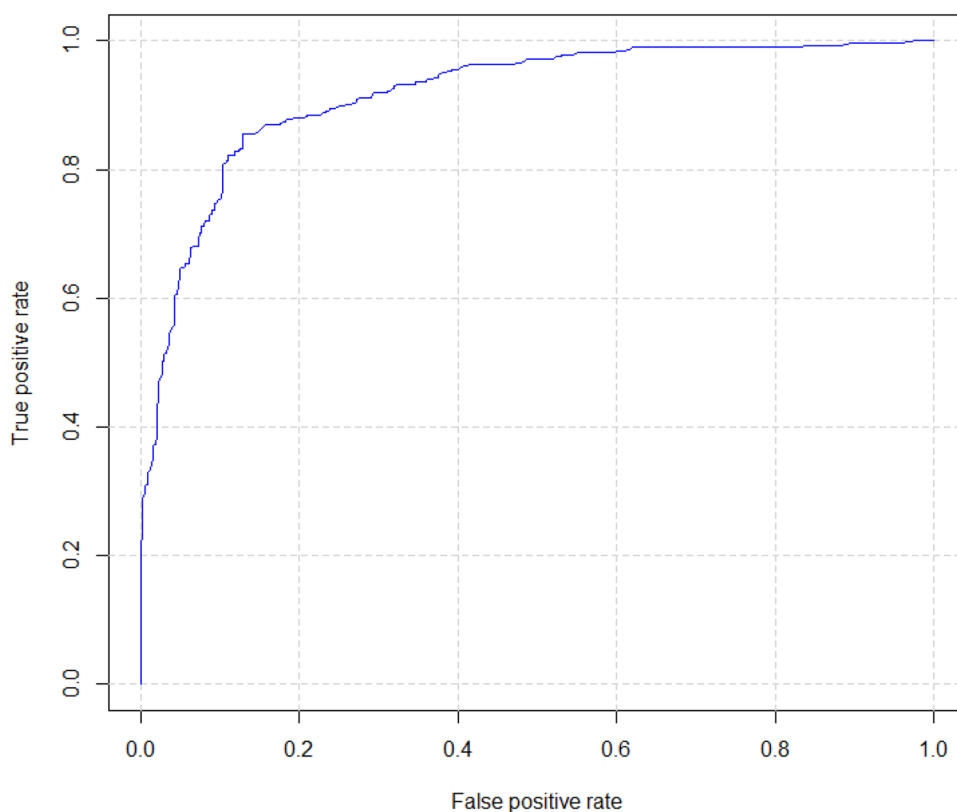
S No.	Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	MCC	F1 score
1	SVM (set A) (C=2.4 & sigma=0.1)	82.01	77.54	85.84	0.63	0.79
2	RF (set B) (mtry=5)	84.17	77.80	89.66	0.68	0.81
3	RF (set C) (mtry = 13)	85.38	82.76	87.64	0.70	0.83

Highlighted values indicate the models selected

Table 5 Optimal cut points of top 2D descriptors

Predictor	Direction	Optimal cut point	Accuracy	Sensitivity	Specificity	Sensitivity + Specificity
maxssCH2	≥	0.608648333	0.640902	0.709042	0.585223	1.294265193
minHssNH	≥	0.157765692	0.612721	0.606983	0.61741	1.224393378
SaasC	≥	0.629098687	0.586554	0.66607	0.52158	1.187649932
minssCH2	≥	0.607225623	0.574074	0.784244	0.402341	1.186584402

Fig. 4 ROC curve of best model (random forest trained using fingerprints) for prediction on external set



Out of 828 molecules from the external set 21 molecules were outside the applicability domain. For 21 molecules which are outside the applicability domain no information rate (NIR) is 61.90%. An accuracy of 76.19% was obtained for these molecules. The sensitivity and specificity were 62.50% and 84.62%, respectively. The P value for alternate hypothesis, i.e., “accuracy is greater than NIR” is equal to 0.1293. This shows that the performance of the final model is comparatively poor for molecules outside the applicability domain. After removing these 21 molecules, there was marginal improvement in accuracy of the model (85.63% as compared to previous accuracy of 85.38%). Also, the sensitivity and specificity were improved a little (83.20% sensitivity and 87.73% specificity). The results are not that much changed because out of 828 only 21 molecules were outside the applicability domain.

Descriptor analysis

Two types of descriptors are used in this work, i.e., continuous 2D descriptors (set A and set B) and Binary fingerprints (set C). Set A and set B have a total of 24 features in common. To analyze the effect of individual descriptors on the activity of molecules, their cut points were calculated with the help of cutpoint package in R language. Results of optimal cut point calculations are given in the supplementary file, and results of top 4 descriptors are given in Table 5. For

analysis of fingerprints, varImp function of caret package was used. Bit 128 of MACCS key was identified as the most important one which corresponds to two CH_2 groups separated by a chain of 2 or more atoms. Bit 24 of estate fingerprint was on second position which corresponds to secondary amine group. Similar type of 2D descriptors was observed as top 2 in optimal cut point calculation, i.e., maxssCH2 (Maximum atom-type E-State: $-\text{CH}_2-$) and minHssNH (Minimum atom-type H E-State: $-\text{NH}-$). This confirms that there is a definite correlation between these descriptors and activity of molecules. Other than these descriptors SaasC (Sum of atom-type E-State: $:\text{C}-$), minssCH2 (Minimum atom-type E-State: $-\text{CH}_2-$), bit 104 of MACCS key (aliphatic CH_2 group) and bit 18 of estate fingerprints (C atom common between two aromatic rings) were also identified as more important as compared to other descriptors.

In Figs. 5, 6, density plots of descriptors best 2D descriptors, i.e., maxssCH2, minHssNH, minssCH2 and SaasC are given. From the density plots, it is evident that the majority of inhibitors or active molecules are above the cutoff line and majority of inactive molecules are below the cutoff line for these descriptors. In case of fingerprints, bit 128 of MACCS key is present in 69.82% of inhibitors and 45.64% of non-inhibitors, while bit 24 of estate fingerprints is present in 60.25% of inhibitors and 37.89% of non-inhibitors. This further proves that there is a definite correlation between selected descriptors and activity of molecules toward acetylcholinesterase enzyme. As mentioned

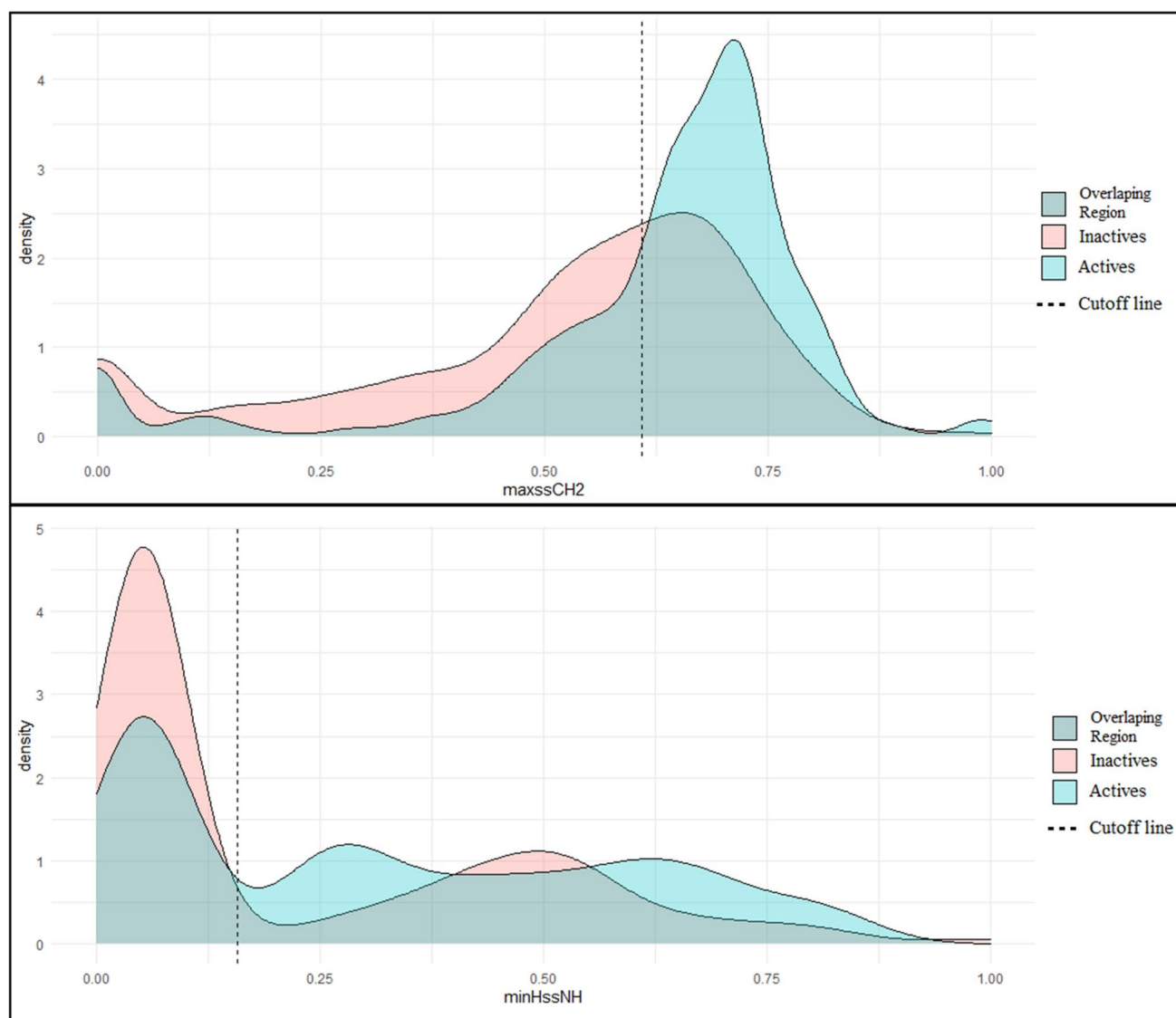


Fig. 5 Density plot of descriptors maxssCH2 and minHssNH (dashed line indicates the cutoff value determined using cut point calculations)

earlier bit 128 of MACCS key corresponds to two CH₂ groups separated by a chain of 2 or more atoms and bit 24 of estate fingerprints corresponds to secondary amine group, which suggests that AChE inhibitors have a flexible chain and nitrogen atom in their structure. Similar types of features are present also in the structure of acetylcholine (natural substrate of acetylcholinesterase). Thus, these types of molecules have a very good chance of binding in the active site of acetylcholinesterase.

Conclusion

In this study, various models were built with descriptors and fingerprints calculated using PaDEL and rcdk, respectively, with different permutation and combination

of various machine learning algorithms (SVM, RF, *k*-NN) and feature selection methods (Boruta and RFE). It was found that model built using fingerprints based on random forest algorithm give better results compared to the other models. The descriptor analysis shows that maxssCH₂, minHssNH, SaasC, minssCH₂, bit 128 MACCS key, bit 104 MACCS key, bit 24 estate fingerprint and bit 18 estate fingerprints are the most important descriptors, which strongly correlates with activity of molecules. Finally, the best overall model was used to build a shiny app which can be used for identification plausible hits from a library of lead molecules. As compared to earlier published models for classification of AChE inhibitors, this work uses much larger dataset for training of models. The developed model is available at <http://14.139.57.41/achepredictor/>

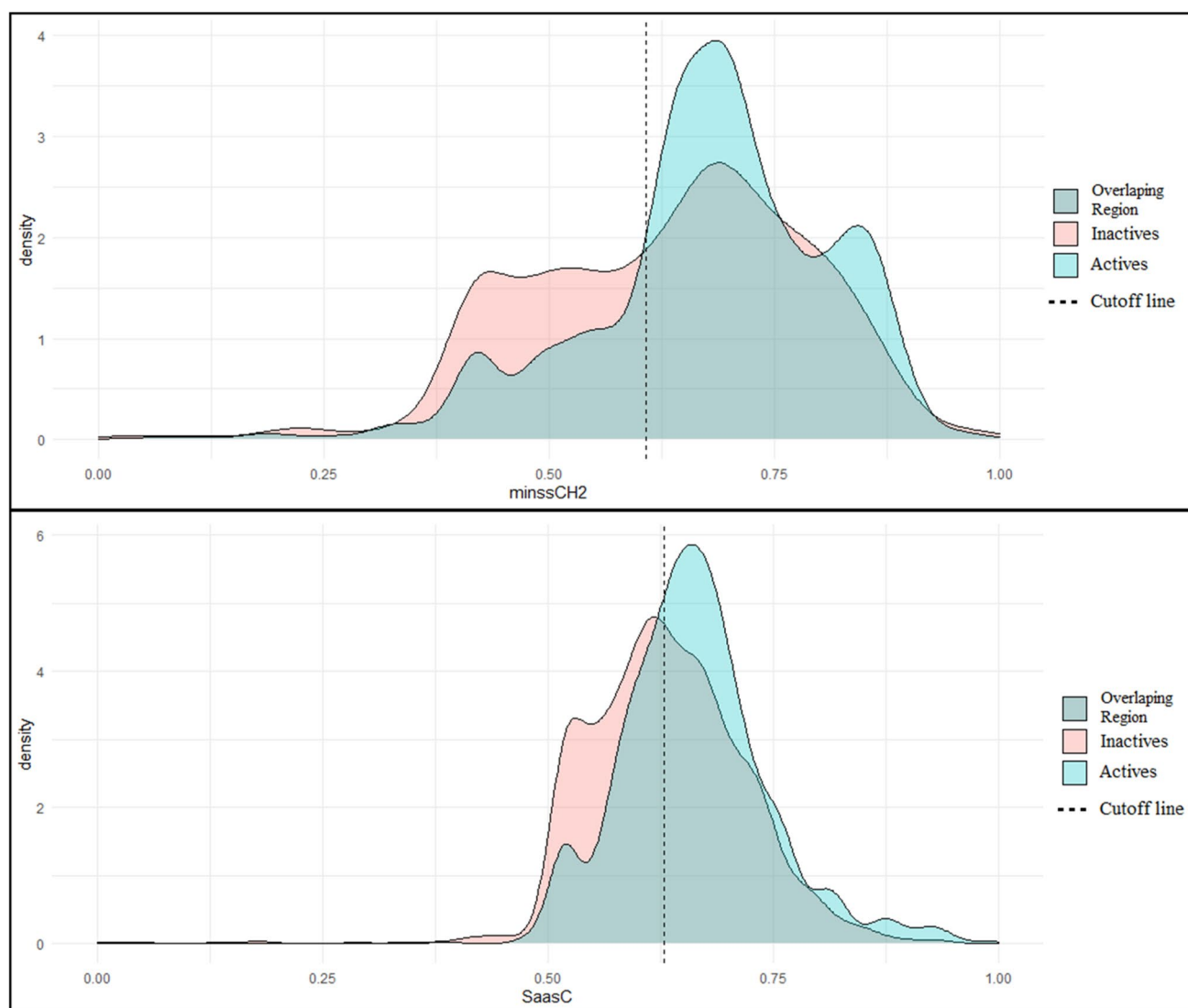


Fig. 6 Density plots of descriptors minssCH2 and SaasC (dashed line indicates the cutoff value determined using cut point calculations)

as downloadable standalone application as well as web application.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11030-021-10223-5>.

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